



ARiS

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Module: PDE3823 Major Project and Professional Practice

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Introduction

Smart Accessibilities. Better Cities. Better Future

- Urban Accessibility is a major challenge in modern cities in India.
- existing methods rely on manual surveys and static rules.
- Lack of real-time, data-driven decision-making tools.
- Need for intelligent systems to assess and prioritise accessibility issues.

Proposed Solution:

- AI-Driven Urban Accessibility Risk Mapping & Prediction System (ARiS)





Problem Statement

01

Accessibility issues are not identified systematically.

02

Current methods are manual, time consuming, and inconsistent.

03

No unified system to combine multiple accessibility factors.

04

Lack of predictive insights and prioritization tools.



Objectives

Primary Objective

Develop an **intelligent** system to assess and predict **accessibility risk**.

Secondary Objectives

- 01 Analyze key factors (width, slope, curb ramp, traffic, lighting, and surface).
- 02 Predict risk using machine learning and provide confidence and explainability of predictions.
- 03 Evaluate Accessibility compliance using rule-based logic.
- 04 Enable Scenario Simulation for improvements and predict accessibility risk for new/unseen scenarios.
- 05 Rank areas using a Priority Intervention Engine.
- 06 Develop a City Planning Dashboard (Budget Optimizer) for resource allocation

Project Scope

What's included in the system

- Accessibility risk prediction (ML-based).
- Compliance evaluation (rule-based).
- Confidence & Explainability layer.
- Scenario simulation (what-if analysis).
- Priority Intervention Engine (ranking urgency).
- Interactive dashboard (map-based visualization).
- Prediction for new/unseen scenarios.
- Model comparison dashboard (Rule vs ML).
- City Planning Dashboard (Budget Optimizer).

What's not included in the system

- Real-time IoT or sensor integration.
- Full city-scale deployment.
- Government policy enforcement.
- Live traffic / real-time data feeds.

Target Users

- Urban planners & City authorities.
- Smart city developers.
- Accessibility consultants.
- Infrastructure decision-makers.
- End beneficiaries: elderly and differently-abled individuals.

System Overview & Architecture



Literature Review

Approach	Limitations
Rule-Based	Not flexible
GIS Systems	No Prediction
ML models	No flexibility
Proposed System	Integrated Solution

Tools & Technologies

Frontend

- React (UI development)
- Leaflet (Map Visualization)
- Tailwind CSS (Styling)

Backend

- FastAPI (API Development)
- Python (Core Logic)

Machine Learning

- Scikit-Learn (Random Forest Model).
- Pandas & NumPy (Data processing).

Other Tools

- Joblib (Model Saving/Loading).
- LocalStorage (Scenario Persistence).

KEY FEATURES

#	FEATURE	WHAT IT DOES
01	 AI-Based Risk Prediction	Predicts accessibility risk using Machine Learning
02	 Rule-Based Compliance Check	Evaluates adherence to accessibility standards
03	 Explainability (XAI Layer)	Shows why a segment is high or low risk
04	 Confidence & Uncertainty Layer	Indicates reliability of predictions
05	 Scenario Simulator (What-if Analysis)	Tests impact of infrastructure improvements
06	 Priority Intervention Engine	Ranks areas based on urgency
07	 Interactive Map Dashboard	Visualises risk, compliance, and priority
08	 Model Comparison Dashboard	Compares rule-based and ML-based outputs
09	 Accessibility Risk Prediction for New Areas	Predicts risk for new or unseen scenarios
10	 City Planning Dashboard / Budget Optimizer	Supports budget-based intervention planning
11	 Recommendation Support	Provides intervention-focused planning insights
12	 End-to-End Integration	All features integrated into a unified system

Key Features

Results & Output



The screenshot displays the ARiS web application interface. At the top, the title 'ARiS - Accessibility Risk Mapping and Prediction Intelligence System' is shown, with a 'Business' label and a 'Backend status: ok - AURaMP ...' indicator. The main navigation bar includes 'Map & Block View', 'Model Insights', 'Model Comparison', and 'City Planning Dashboard'. The left sidebar contains filters for 'VIEW MODE' (Rule-based, ML-based), 'RISK FILTER' (All, High, Medium, Low), 'COMPLIANCE FILTER' (Show all blocks), and 'PRIORITY INTERVENTION ENGINE' (Ward / Zone: Ward-1, Run Priority Engine for Ward). The central map area shows a street view with a red play button overlay. The right panel, titled 'Fort Area Lane', displays 'AI PREDICTION MODE' results: Rule-based (0.75, HIGH) and ML-predicted (0.95, HIGH, 98.8% confidence). Below this is a 'CPWD Compliance Scorecard' showing 2/5 compliance (40%) with a list of factors: 'Clear width > 1.2m' (Non-compliant, Value: 1.10 m), 'Slope ≤ 8.33% (1:12)' (Compliant, Value: 6.00 %), 'Curb ramp present at crossing' (Non-compliant, Value: no), 'Surface firm & even (smooth)' (Non-compliant, Value: uneven), and 'Adequate pedestrian lighting' (Compliant, Value: good). A reference to 'MoHUA/CPWD Harmonised Guidelines (Universal Accessibility) - min clear width 1.2m, slope reference 1:12' is provided. The bottom right corner features a 'Watch on YouTube' button.

Metric	Value
Total Blocks	155
Avg Compliance	43%
High Risk	104
Medium Risk	20
Low Risk	31

Youtube link: <https://youtu.be/yHGk-SPQWcl>

Challenges Faced

CHALLENGES FACED

S.No.	Challenge	Description
1	 Data Availability & Quality	Limited real-world accessibility datasets
2	 Balancing ML & Rule-Based Logic	Ensuring consistency between both approaches
3	 Model Explainability	Making predictions understandable (XAI)
4	 Feature Selection	Choosing relevant factors for accurate prediction
5	 Integration Complexity	Connecting frontend, backend, and ML model
6	 Weight Tuning (Priority Engine)	Assigning meaningful importance to factors



Future Plans

- Real-Time Data Integration such as IoT sensors, live traffic, GIS APIs.
- Large-Scale Deployment for city-wide implementation.
- Advanced ML models such as Deep Learning & improved accuracy.
- More Features such as pedestrian flow, weather, and time-based factors.
- Mobile Application for On-field accessibility assessment.
- Adaptive Priority Engine for Dynamic weight adjustment based on data.



Conclusion

- Developed an AI-driven accessibility analysis system.
- Successfully combined ML + rule-based + explainability.
- Provided risk prediction, compliance, and prioritisation.
- Enabled data-driven decision-making for urban planning.
- Demonstrated practical and scalable solution potential.

THANK YOU!!!